Identity and Access Management (IAM) in Cloud Environments

Authentication, authorization and enterprise implementation

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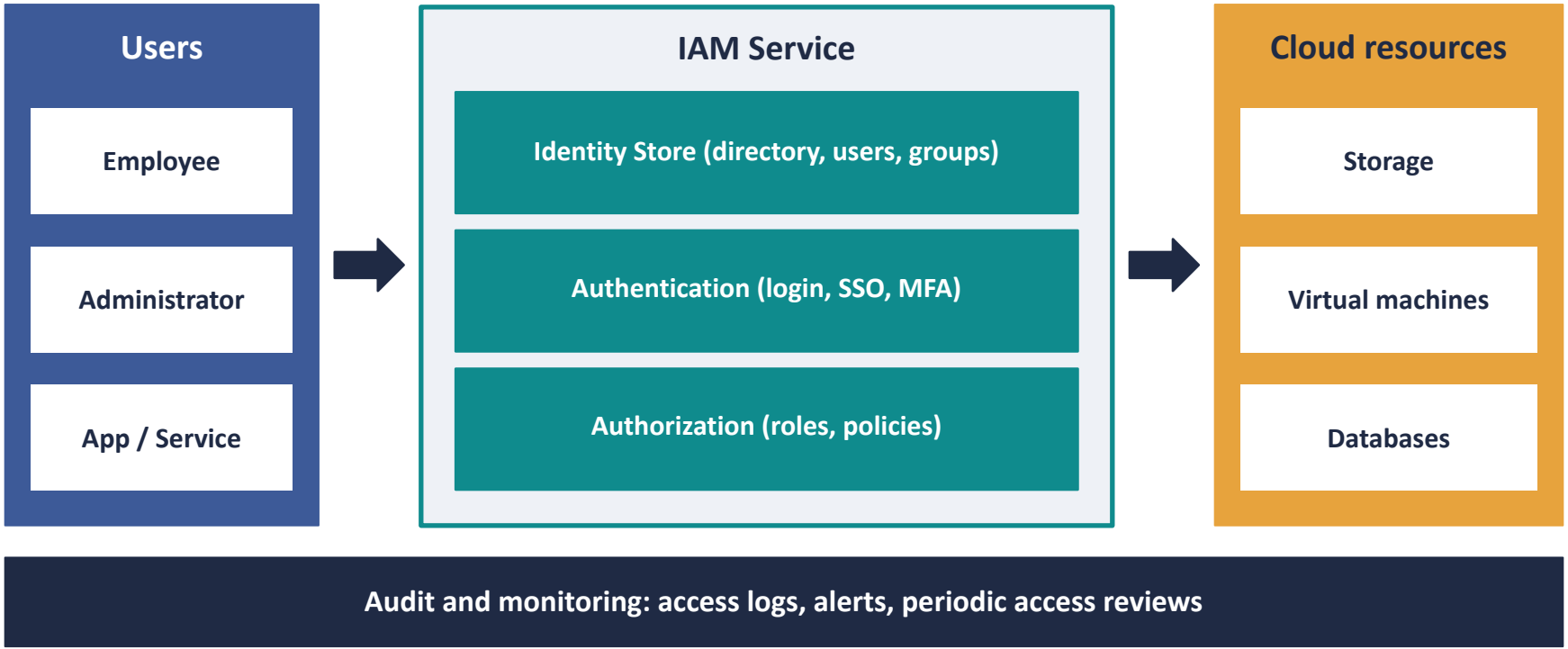
Introduction

- Identity and Access Management manages user identities and access to cloud resources.
- It ensures the right user gets access to the right resource at the right time.
- Authentication verifies the identity of the user.
- Authorization decides what resources and actions the user is allowed to access.
- It helps protect cloud data, applications, and services from unauthorized access.

Core concepts of IAM

Concept	Meaning	Example
Authentication (AuthN)	Proves identity	Password + OTP
Authorization (AuthZ)	Grants permissions	Read-only access to storage
SSO	One login for many apps	Sign in once to email and HR portal
MFA	Two or more proofs of identity	Password + phone prompt
RBAC	Permissions given by role	"Developer" role, "Auditor" role
Least privilege	Only the access needed for the job	No admin rights for a tester

System architecture



Mechanisms and standards

Mechanism	Purpose	Where used
SAML 2.0	Federated SSO using XML assertions	Enterprise web apps
OAuth 2.0	Delegated authorization with tokens	"Allow this app to access my data"
OpenID Connect	Identity layer on top of OAuth 2.0	Social / "Sign in with" logins
Kerberos / LDAP	Ticket-based login and directory lookup	Corporate networks, Active Directory
MFA / biometrics	Stronger proof of identity	Banking, admin consoles

Security benefits and industry relevance

Security benefits

- Prevents unauthorized access
- Limits damage from stolen credentials
- Full audit trail

Industry use

- Banking: protect transactions
- Healthcare: patient records
- IT and e-commerce: staff and customer logins

Compliance

- GDPR
- HIPAA
- PCI DSS
- SOX

Conclusion

- IAM decides who can access which cloud resource, and under what conditions
- Authentication proves identity; authorization grants permissions
- Core practices: MFA, SSO, RBAC and least privilege
- Architecture: identity store, authentication, authorization, audit logs
- IAM is essential for security and compliance in every industry using the cloud

Thank You